A NEGATIVE ANSWER TO ORDOWSKI'S SECOND QUESTION

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Abstract. Let $E(n)$ be the largest exponent in the prime factorization of n. Thomas Ordowski asked whether an odd integer n with $E(n) > 1$ can satisfy $n \equiv E(n) \pmod{\lambda(n)}$, where λ is Carmichael's function. We give a negative answer. If an odd prime p occurs in n to the maximal exponent e, the proposed congruence forces $p^{e-1} \mid e$, whereas $p^{e-1} > e$. We also give an explicit base for which the equivalent universal power congruence fails, and characterize all nonsquarefree solutions: they have the form $4m$ or $16m$, with m odd and squarefree and with specified divisibility conditions on its prime factors. The nonexistence theorem is formally verified in Lean using Mathlib.

1. Introduction

For a positive integer $n > 1$, write

$$n = \prod_{i=1}^{r} p_i^{f_i}, \qquad E(n) = \max_{1 \le i \le r} f_i, \tag{1.1}$$

where the p_i are distinct primes, and set $E(1) = 0$. The function E is sequence A051903 in the On-Line Encyclopedia of Integer Sequences. In a comment dated December 2, 2019, Thomas Ordowski asked whether there is an odd positive integer n such that

$$E(n) > 1, \qquad n \equiv E(n) \pmod{\lambda(n)}, \tag{1.2}$$

where $\lambda(n)$ is the exponent of the group $(\mathbb{Z}/n\mathbb{Z})^\times$ [4]. This is the second of the three questions in that comment.

The same question is recorded in the Formal Conjectures repository as `OeisA51903.conjecture2` [1], in the equivalent form

$$b^n \equiv b^{E(n)} \pmod{n} \qquad \text{for every } b \in \mathbb{Z}_{\ge 0}. \tag{1.3}$$

The equivalence is proved in Proposition 2.1; it includes bases that are not relatively prime to n.

Theorem 1.1. *There is no odd positive integer n for which $E(n) > 1$ and $\lambda(n) \mid n - E(n)$. Equivalently, every odd positive integer with $E(n) > 1$ fails* (1.3)*.*

The proof uses only the prime-power formula for Carmichael's function. At a prime that attains the largest exponent in n, the proposed congruence forces a prime power to divide a smaller positive integer. Section 3 makes the obstruction explicit: if $p^{E(n)} \parallel n$, then the base $b = 1 + p$ violates (1.3). Section 4 gives necessary and sufficient conditions for every nonsquarefree solution, including the even ones. In particular, it establishes the suggested structure of sequence A327295 [5]. Section 5 describes the Lean verification of Theorem 1.1.

Throughout, $v_p(a)$ denotes the exponent of a prime p in a nonzero integer a. The notation $p^f \parallel n$ means $v_p(n) = f$. All divisibility statements are statements about integers.

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2. The Carmichael criterion and the local obstruction

We use the standard formulas

$$\lambda(n) = \operatorname{lcm}_{q^f \| n} \lambda(q^f), \qquad \lambda(q^f) = q^{f-1}(q-1) \quad (q \text{ odd}), \tag{2.1}$$

and

$$\lambda(2) = 1, \qquad \lambda(4) = 2, \qquad \lambda(2^f) = 2^{f-2} \quad (f \geq 3). \tag{2.2}$$

These follow from the Chinese remainder theorem and the structure of unit groups modulo prime powers. The corresponding formal results are available in Mathlib's development of Carmichael's function [3].

Proposition 2.1. *Let $n > 1$ and put $e = E(n)$. Then*

$$\big(\forall b \in \mathbb{Z}_{\geq 0},\ b^n \equiv b^e \pmod{n}\big) \quad \Longleftrightarrow \quad \lambda(n) \mid n - e. \tag{2.3}$$

Proof. First, $1 \leq e < n$: a prime power p^e divides n, and $e < 2^e \leq p^e \leq n$. Suppose the left side of (2.3) holds. For each unit $u \in (\mathbb{Z}/n\mathbb{Z})^\times$, cancel u^e from $u^n = u^e$ to obtain $u^{n-e} = 1$. The exponent of this finite group therefore divides $n - e$.

Conversely, suppose $\lambda(n) \mid n - e$ and fix $b \geq 0$. For each $q^f \| n$, if $q \nmid b$, then $\lambda(q^f) \mid n - e$ gives $b^{n-e} \equiv 1 \pmod{q^f}$ and hence $b^n \equiv b^e \pmod{q^f}$. If $q \mid b$, both b^n and b^e vanish modulo q^f, because $n > e \geq f$. Thus the congruence holds modulo every prime-power factor of n, and the Chinese remainder theorem proves it modulo n. □

Lemma 2.2. *Suppose $n > 1$, $e = E(n) \geq 2$, and $\lambda(n) \mid n - e$. No odd prime occurs in n to exponent e.*

Proof. If p is an odd prime and $p^e \| n$, then (2.1) gives $p^{e-1} \mid \lambda(n)$, so $p^{e-1} \mid n - e$. Since $p^{e-1} \mid n$ as well, it follows that

$$p^{e-1} \mid e. \tag{2.4}$$

But

$$p^{e-1} \geq 3^{e-1} > e \qquad (e \geq 2), \tag{2.5}$$

contradicting (2.4). For completeness, the strict inequality follows by induction: $3 > 2$, and $3^{e-1} > e$ implies $3^e > 3e > e + 1$. □

Proof of Theorem 1.1. For an odd n with $E(n) \geq 2$, at least one prime attains the maximum in (1.1), and every prime divisor of n is odd. Lemma 2.2 excludes $\lambda(n) \mid n - E(n)$. Proposition 2.1 gives the equivalent assertion about (1.3). □

3. An explicit base witnessing failure

The following elementary valuation identity identifies a base that fails for every odd candidate. We include the binomial proof to make this part of the argument self-contained.

Lemma 3.1. *For an odd prime p and a positive integer d,*

$$v_p\big((1+p)^d - 1\big) = 1 + v_p(d). \tag{3.1}$$

Proof. We first show that for each $t \geq 0$ there is an integer c_t with

$$(1+p)^{p^t} = 1 + p^{t+1} c_t, \qquad c_t \equiv 1 \pmod{p}. \tag{3.2}$$

For $t = 0$, take $c_0 = 1$. Raise (3.2) to the pth power. The linear term is $p^{t+2} c_t$. For $2 \leq j \leq p - 1$, the term $\binom{p}{j} p^{j(t+1)} c_t^j$ is divisible by p^{t+3}, since $p \mid \binom{p}{j}$ and $1 + j(t+1) \geq t + 3$. The last term is also divisible by p^{t+3}, because $p(t+1) \geq t + 3$ for $p \geq 3$. Thus (3.2) holds for $t + 1$, with $c_{t+1} \equiv c_t \equiv 1 \pmod{p}$.

Now write $d = p^t u$, where $u \geq 1$ and $p \nmid u$. Raising (3.2) to the uth power gives

$$(1+p)^d \equiv 1 + u c_t p^{t+1} \pmod{p^{t+2}}.$$

Indeed, all terms of degree at least two in $p^{t+1}c_t$ are divisible by p^{2t+2}, and $2t+2 \geq t+2$. Since $p \nmid uc_t$, the claimed valuation follows. □

Theorem 3.2. *Let n be an odd positive integer with $e = E(n) \geq 2$, and choose a prime p such that $p^e \parallel n$. For $b = 1 + p$,*

$$v_p(b^n - b^e) = 1 + v_p(e) < e. \tag{3.3}$$

In particular, $b^n \not\equiv b^e \pmod{n}$.

Proof. Set $t = v_p(e)$. By (2.5), $e < p^{e-1}$, so $t < e - 1$. Since $v_p(n) = e > t = v_p(e)$, the elementary valuation rule for a difference with unequal valuations gives $v_p(n - e) = t$. Also $n \geq p^e > e$, so $n - e$ is positive. The integer b is prime to p; hence Lemma 3.1 yields

$$v_p(b^n - b^e) = v_p\left(b^{n-e} - 1\right) = 1 + t < e.$$

Thus $p^e \nmid b^n - b^e$. As $p^e \mid n$, the congruence fails modulo n. □

Remark 3.3. The chosen base need not be relatively prime to every prime divisor of n. For example, $n = 75$ has $E(n) = 2$, and the choice $p = 5$ gives $b = 6$, which shares the factor 3 with n. Here $6^{75} \equiv 1 \pmod{25}$, whereas $6^2 \equiv 11 \pmod{25}$. Failure at a single prime-power factor suffices.

4. Classification of the nonsquarefree solutions

Removing the requirement that n be odd gives a complete local criterion. The possible exponents of 2 are especially restricted.

Theorem 4.1. *Let $n > 1$ be nonsquarefree. The universal congruence* (1.3) *holds if and only if one of the following applies:*

(i) *$n = 4m$, where m is odd and squarefree, and $q - 1 \mid 4m - 2$ for every prime $q \mid m$;*

(ii) *$n = 16m$, where m is odd and squarefree, and $q - 1 \mid 16m - 4$ for every prime $q \mid m$.*

In both cases $m = 1$ is allowed, with the prime conditions vacuous.

Proof. Suppose (1.3) holds, and set $e = E(n) \geq 2$. Proposition 2.1 gives $\lambda(n) \mid n - e$. By Lemma 2.2, an odd prime cannot attain the maximal exponent, so $2^e \parallel n$. If $e \geq 3$, formula (2.2) implies $2^{e-2} \mid n - e$. Because $2^{e-2} \mid n$, we obtain

$$2^{e-2} \mid e. \tag{4.1}$$

The case $e = 3$ is impossible, while $e = 4$ satisfies this necessary condition. For $e \geq 5$, induction from $2^3 > 5$ gives $2^{e-2} > e$, which contradicts (4.1). Thus $e \in \{2, 4\}$.

For any odd prime power $q^f \parallel n$, (2.1) gives $q^{f-1} \mid \lambda(n) \mid n - e$. As $q^{f-1} \mid n$, it follows that $q^{f-1} \mid e$. Since e is either 2 or 4 and q is odd, we must have $f = 1$. Consequently $n = 2^e m$ with m odd and squarefree. The remaining factors of $\lambda(n)$ give $q - 1 \mid n - e$ for each $q \mid m$, as claimed.

Conversely, either description gives $E(n) = e$, with $e = 2$ or 4. Each odd prime divisor q contributes $\lambda(q) = q - 1$ to the least common multiple in (2.1), and this divides $n - e$ by hypothesis. The contribution from 2^e is 2 when $e = 2$ and 4 when $e = 4$; it too divides $n - e$. Thus $\lambda(n) \mid n - e$, and Proposition 2.1 proves (1.3). □

Corollary 4.2. *Every term n of OEIS A327295 is even and satisfies*

$$E(n) = v_2(n) \in \{2, 4\}, \qquad n/2^{E(n)} \text{ is squarefree.}$$

Proof. The defining conditions of A327295 are $E(n) > 1$ and $\lambda(n) \mid n - E(n)$ [5]. Apply Proposition 2.1 and Theorem 4.1. □

For example, the first description includes 4, 12, and 132; the second includes 16, 48, 80, and 112. The classification provides a necessary and sufficient arithmetic test. It does not settle whether there are infinitely many such integers. Nor does the nonexistence theorem settle Ordowski's separate question about the single fixed base 2 [4].

5. Formal verification and computational checks

The accompanying file `OrdowskiConjecture2.lean` proves the nonexistence assertion of Theorem 1.1 in Lean 4.33.1 using Mathlib [2]. It imports only `Mathlib`; all additional lemmas used in the proof are included in the file.

The definition `OeisA51903.a` is copied from the Formal Conjectures statement. It forms the list of prime factors of n, counts the multiplicity of each entry, and takes the maximum, with default value zero. Thus it is exactly the function E for positive integers. The final declaration, `OeisA51903.conjecture2`, proves

$$\text{False} \longleftrightarrow \exists n \in \mathbb{N}, \quad n \text{ odd}, \quad 1 < a(n), \quad \forall b \in \mathbb{N},\ b^n \equiv b^{a(n)} \pmod{n}. \tag{5.1}$$

Here $\mathbb{N}$ includes zero; oddness excludes that case. The negative answer resolves the answer placeholder in the original declaration, without altering its remaining hypotheses or conclusion.

The formal proof follows Section 2. It extracts a prime attaining the maximum multiplicity, passes from the universal congruence to the exponent of the unit group, and uses the odd prime-power formula for Carmichael's function to derive $p^{e-1} \mid e$. The strict inequality $e < p^{e-1}$ is proved by induction. A separate audit checks the resulting theorem against the original repository definition and statement.

The file was compiled with warnings treated as errors. Lean's axiom report for the final theorem lists only `propext`, `Classical.choice`, and `Quot.sound`; the proof has no unproved placeholders or additional assumed mathematical results. The checked Mathlib revision is

`0df444a360eaa60ab8c11dca51a86af692955474`.

Within the supplied Lean project, the verification command is

```
lake env lean -DwarningAsError=true OrdowskiConjecture2.lean
```

The explicit valuation result and the classification in Sections 3 and 4 are proved in this paper; they are not asserted as additional theorems in the accompanying Lean file.

As an independent check on the formulas, exact integer computations verify the classification for every $1 \le n \le 10^6$ and the explicit witness for all 94 714 odd nonsquarefree integers in that range. There are 38 nonsquarefree solutions up to 10^6. Direct modular exponentiation checks all residue bases for each passing candidate up to 2000, stopping at a failing base for the other candidates; this tests 290 812 bases in total. These finite computations are consistency checks, and the proofs above establish the unrestricted mathematical statements.

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